

LSI-lite: A Composition Analysis Tool (with Color Telemetry)

Profiled Composition Structural Exploration (Δx , r_v , ρ_r)

What It Does

LSI-lite measures how an image **behaves under compositional structure** using three primitives: Δx (off-center gravity), r_v (void ratio), ρ_r (rupture/mark energy) and tells you if it sits within intended bands for its class. It's built to study **stability and failure** as you iterate, not to crown winners.

- **Balance** (Δx): How far the visual center is from the geometric center
- **Void** (r_v): The ratio of empty space to filled space
- **Detail** (ρ_r): The amount of edge energy and texture density in key areas

It combines these measurements into a 0-100 score and tells you whether an image "passes" basic compositional criteria structural exploration test.

→ This version now includes: **Color telemetry (advisory only)**. It logs a color-aware void mask and a luminance-only balance read to explain divergences. These **never gate** acceptance; they only trigger a one-line **Color audit** when the gap is material.

Why researchers use it (not FID/CLIP/SSIM).

Resemblance and caption metrics can't answer: *will this composition hold* when pushed? LSI-lite is a **composition structural exploration test** and a quick, profiled gate you can log across Baseline → Pressure → Collapse-trigger runs.

The Three Measurements

Δx (Delta-x): Off-Center Balance

- Finds the visual center of mass using edge detection
- Measures how far it is from the geometric center
- Range: 0.0 (perfectly centered) to 1.0 (at edge)
- **Why it matters:** Dead-center compositions often feel static

r_v (Void Ratio): Empty Space

- Calculates the percentage of background vs foreground
- Higher values = more empty space
- Range: 0.0 (completely filled) to 1.0 (mostly empty)
- **Why it matters:** Negative space creates visual breathing room
- → Color note: when color helps isolate subject/background, we also log a color-derived void read for comparison (see Color telemetry below).

ρ_r (Rho-r): Rupture Energy

- Measures texture and detail density around the main subject
- Uses Laplacian edge detection over a "halo" around the subject
- Range: 0.0 (smooth) to 1.0 (highly detailed)
- **Why it matters:** Controls visual complexity and focus

Three Profiles

Different types of images need different compositional rules, this lite version covers:

Figure_Default

- **For:** Portraits, figure drawings, character art
- **Emphasis:** Balance (45%), then density (35%), then detail (20%)
- **Expects:** Moderate off-centering, balanced space usage, controlled detail

MarkMaking_Expressive

- **For:** Sketches, expressive artwork, gestural drawings
- **Emphasis:** More balanced weights (40%/30%/30%)
- **Expects:** Looser centering rules, higher texture tolerance

Landscape

- **For:** Landscape photos and paintings
- **Emphasis:** Density matters most (40%), then balance (35%), then detail (25%)
- **Special feature:** Detects water reflections and adjusts balance measurement accordingly

- → *Reflection check*: for strong water reflections, balance may be read on a shallower top window to avoid mirroring bias.

Gate rule (accept/reject).

- Combine the three measurements into LSI_lite_100 (0–100). **Accept** only if LSI_lite_100 ≥ 55 **and** no bands are **RED**; otherwise **reject** and name the limiting primitive (**priority_knob**).

Telemetry (advisory only)

Color audit thresholds (non-gating):

- Emit a Color audit only if the color-based void read differs **meaningfully** from gray, or the luminance-only balance read shifts **meaningfully** from gray.
- Rule of thumb: log an audit when the gap is **noticeable**; suppress when the reads agree. (*Engineering note for later: ≈0.15 void gap or ≈0.10 balance gap, or gray void sits right on a guard ±0.02.*)

How Scoring Works

1. **Measure**: Calculate all three primitives (Δx , r_v , p_r)
2. **Check bands**: Each primitive has an acceptable range - values outside are flagged "RED"
3. **Score**: Combine the three values using weighted geometric mean
4. **Gate**: Image "passes" if score ≥ 55 AND no RED flags
5. **(If warranted) Color audit**. If color-based reads diverge, add one line under the result. No change to accept/reject.

Example Output

What a single run gives you (at a glance).

- **Grade card**: profile • $\Delta x/r_v/p_r$ (0–1) • per-band OK/RED • LSI_lite_100 • **accepted** True/False • **priority_knob** (+ suggested raise/lower)
- **Stamp**: ACCEPT / REJECT / UNKNOWN — reason_for_fail=<enum>
- **Color audit** (*only when thresholds trip*):
- voids under color are lighter than gray; confirm subject mask / glare
- **Exports**: a composite PNG + one CSV/JSON row (fields below).

New fields (advisory):

- r_v _color_mask — color-derived void read
- dx _color_L — luminance-only balance read
- Δr_v — difference between color void and gray void (*for engineers*)
- $\Delta \Delta x$ — difference between luminance balance and gray balance (*for engineers*)
- reason_for_fail — short enum: band_red, lsi_gate, unknown_input, calc_error, roi_mismatch, mask_empty
- color_audit_badge — **one line**; present only when an audit fires

What it is *not*.

Not aesthetic judgment. "Accepted" means "within intended structural bands," not "good art." Profiles reflect specific design intent; many excellent images will fail on purpose.

What This Tool Is Good For

- **Art education**: Consistent feedback on basic composition principles
- **Batch analysis**: Evaluate large sets of images quickly
- **AI research**: Study how different generation methods handle composition
- **Quality filtering**: Screen images for basic structural soundness

Important Limitations

Not Universal

- Built around Western compositional conventions
- Many excellent artworks will fail these tests intentionally
- Abstract art, experimental compositions may be penalized unfairly

Not Aesthetic Judgment

- "Accepted" means "follows basic rules," not "good art"
- Measures structure, not beauty, emotion, or meaning

- Doesn't evaluate color harmony/palette or meaning; ****color telemetry is diagnostic only**** (masking & luminance separation), not an aesthetic score.

Current Scope

- Three primitives can't capture all compositional elements
- Profile bands are designer intuition, not empirically validated
- Works best on clear subject/background images

Technical Notes

- Handles EXIF rotation automatically
- Resizes to 1536px longest side for consistency
- Uses bilateral filtering and morphological operations for robustness
- Fallback handling for edge cases (no foreground detected, etc.)

"Limits"

Where Color Telemetry Can Mislead (watch-outs):

- Heavy cast lighting or LEDs can fake "void" where none exists.
- Glossy black or neon backgrounds may confuse the color mask.
- Busy fabrics/patterns around a centered subject can nudge the balance read.
(*These are audit triggers, not automatic fails.*)

Research Context

LSI-lite is designed as a research instrument to study compositional stability in AI-generated images. Unlike metrics that measure similarity to training data (FID) or text-image alignment (CLIP), this focuses on structural principles that observe how compositions hold up under iteration or modification. This also logs **color telemetry**, a color-aware void read and a luminance-only balance read, to explain divergences across materials and engines; these notes are **diagnostic only** and never affect acceptance.

The goal is not to replace human artistic judgment, but to provide a systematic way to discuss why some compositions feel more stable than others. LSI-lite reports **what shifts**, not why it's "good"; color telemetry is a compass note, not a verdict.

Artist's Statement

Structural Exploration. Artists don't march images through single images or a linear test; we explore a space of alternatives. Pressure is applied as a contradiction (e.g., torque↑ & permeability↑). The image may collapse, re-cohere, or discover a **new center**. LSI-lite doesn't score "beauty"; it logs whether off-center gravity (Δx), breathing ground (r_v), and mark energy (p_r) support the path you're taking. The guardrails aren't grades, they're **questions**: where does gravity sit, do voids breathe, are marks committed? Exploration ends when form is found or the gap is named.

How LSI fits that story

- **Δx (gravity)** → "Did we escape centering in a way that reads as intentional gravity, not tip?"
 - *Color note (advisory):* when luminance isolates structure better than gray, we log a **luminance-only balance read** for comparison.
- **r_v (voids)** → "Are the alternatives ventilating, or sealing?"
 - *Color note (advisory):* when color separates subject/background, we also log a **color-derived void read**; it's reported only when it materially diverges from gray.
- **p_r (rupture)** → "Is energy committed, or decorative noise?"
- **Gate (≥ 55 , no RED)** → "Is this branch structurally sound *for this profile*?"
 - *Telemetry does not gate.* Color reads may trigger a **one-line Color audit**; acceptance stays with the meter.
- **Priority knob** → "If we pursue this branch, what single nudge makes it truer to itself?"

Non-linear path (what we actually see)

- **Default** → **Collapse** → **Cohesion** → **New Center**
or
- **Default** → **Deviation** → **Poise (hold)** → **Deconstruction (on purpose)**
LSI marks the **state changes**, not a pass/fail morality.

Color Telemetry — Watch-outs

- Heavy casts/LEDs can fake “void.” • Glossy blacks/neons can confuse the mask. • Busy patterns near center can nudge balance.
(These trigger an audit line only; they don’t fail the image.)

*It’s fieldwork in a landscape of alternatives. Each image is a branch; pressure asks whether the branch **finds form** or collapses into defaults (human and model). LSI isn’t judging taste; it’s the start of a **compass** that tells you where structure holds while you deviate.*

This is an experimental tool. Use it to explore patterns in images, not as a definitive judgment of artistic quality.

Appendix:

Methods (how it works, reproducible)

Primitives (what/why/how)

- **Δx — Centroid offset** (off-center gravity / frame tension)
How: bilateral denoise → **Canny(50,100)** → **edge centroid**; **fallback** to foreground centroid if edges empty. If both fail ⇒ **$dx = \text{NaN}$** (mark **unknown**, gate-fail).
- **r_v — Void ratio** (figure-ground breath / negative space)
How: Otsu threshold → largest component → open/close → **$rv = 1 - \text{fill}$** .
- **p_r — Rupture energy** (mark/texture pressure)
How: **Laplacian k=3** → abs → normalize → **mean over subject + halo** (halo fraction per profile).
- **Color telemetry (advisory only)**
How: derive a color-aware foreground mask (LAB k-means, k=3; union of non-background clusters). Using that mask we log:
 - **rv_color_mask** — void ratio from the color mask
 - **dx_color_L** — Δx from the LAB L-channel (luminance)
 Why: explains divergences vs. gray reads; **never gates** acceptance.

Fields are logged even when the audit doesn’t fire; they’re for diagnosis only.

Profiles (guards & masks)

Each profile specifies weights and **band guards** (min..max). **Figure/MarkMaking** use a subject halo (0.08 / 0.12). **Landscape** uses **$\rho_{mask}=\text{full}$** and may apply **Δx ROI** via a reflection probe (top_frac=0.60) *for Δx only*. ** r_v / p_r are always full-frame*.

Color telemetry is **profile-agnostic** (same thresholds for all profiles) and follows the same ROI decision used for Δx (e.g., Landscape reflection probe). It **does not** change band guards or weights.

Scoring

- Per-primitive **Gaussian in-band score** (0–1) from guard range.
- **K_lite** = weighted geometric mean of the three soft scores with **$\epsilon=1e-6$** floor.
- **$LSI_lite_100 = 100 \times K_lite$** .
- **Accept** iff **$LSI_lite_100 \geq 55$** and all bands **OK** and not **unknown**.
- Telemetry is **not** part of **K_lite** and does **not** affect **LSI_lite_100** . It only supports a one-line **Color audit** when warranted.

Quality-of-life diagnostics (for iteration)

Alongside $\Delta x/r_v/p_r$:

$*_center$ (band midpoint) • **$*_to_center$** • **$*_to_center_norm$** (0..1) • **$*_direction$** (“increase/decrease”) • **priority_knob** (which primitive is farthest from center) • **priority_score**.

Also emit: **mask_mode** (“gray” | “color” | “fallback_gray”), **color_space** (“LAB”), **color_status** (“ok” | “fail_mask” | “fail_convert” | …) for quick triage.

Unknown semantics

If edges and foreground both fail: **dx = NaN**, **unknown=True**, bands = RED, gate-fail (this prevents silent “passes” on blank/edgeless frames).

Run loop (single pass)

Compute $\Delta x/r_v/p_r$ once, then compute telemetry (**rv_color_mask**, **dx_color_L**) and deltas:

delta_rv = **rv_color_mask** - **void_ratio**, **delta_dx** = **dx_color_L** - **delta_x**.

Build **color_audit_badge** only if thresholds trip; never recompute/overwrite Δx .

I/O and types

Masks are **uint8**; don't truth-test NumPy arrays (**if arr:**). Use **is None**, **np.any**, **np.all**. Runs local or Colab (no hard pins unless you need exact repro).

Color failures do **not** propagate to acceptance. If the color mask or L-read fails, set **color_status** accordingly and leave **rv_color_mask** / **dx_color_L** as **NaN**; no audit line is emitted (telemetry never gates).

Color telemetry follows the same Δx ROI decision (e.g., Landscape reflection probe) and never alters guards or weights.

What you get: Visualizations & Artifacts

Single-image visualization (3×3 grid)

1. Original (blue = geometric center; red = Δx centroid)
2. Grayscale used for analysis
3. Foreground mask (fill %)
4. Edges (Canny 50/100)
5. Halo mask preview (for p_r)
6. Laplacian energy heatmap (p_r field)
7. **Band gauges** ($\Delta x/r_v/p_r$): guard bar + center tick + measured marker + “increase/decrease”
8. **Gaussian curves** (tiny sparklines per primitive) with current value plotted
9. **Acceptance stamp (ACCEPT / REJECT / UNKNOWN)** + one-line reason for fail
Optional overlays: reflection ROI (Δx only), value histogram.
10. Color mask preview (over gray)
11. Luminance balance line (L-only Δx overlay)
12. **Color audit badge** (only if thresholds trip; one line under the stamp)

Exports:

- **PNG** of the grid
- CSV/JSON row per image with this schema (order flexible):
- **delta_x**, **void_ratio**, **rupture_rho**, **K_lite**, **LSI_lite_100**
- **band_delta_x**, **band_r_v**, **band_rho_r**, **accepted**, **frame**, **profile**
- **dx_inside_band**, **dx_to_center**, **dx_to_center_norm**, **dx_direction**
- **rv_inside_band**, **rv_to_center**, **rv_to_center_norm**, **rv_direction**
- **rho_inside_band**, **rho_to_center**, **rho_to_center_norm**, **rho_direction**
- **priority_knob**, **priority_score**
- **Color telemetry:** **rv_color_mask**, **dx_color_L**, **delta_rv**, **delta_dx**, **mask_mode**, **color_space**, **color_status**
- **UX:** **reason_for_fail** (enum:
band_red|lsi_gate|unknown_input|calc_error|roi_mismatch|mask_empty), **color_audit_badge**
(one line; present only when audit fires)
- **module** — “*engine/ridep*” or “*stabilizer/ridp*” when other modules write rows (leave empty for pure LSI runs).

Small watch-outs (drop near the export section, optional)

Color Telemetry Watch-outs: heavy color casts/LEDs can fake “void”; glossy blacks/neons can confuse masks; dense patterned fabrics near center can nudge the balance read. These trigger an **audit**, not a fail.

Batch visualization (dataset)

- **Score histogram** (+ gate line @55, acceptance rate)
- **Band failure breakdown** (% RED by $dx/r_v/p_r$)
- **Priority-knob distribution** (what limits most, and direction $\uparrow/\downarrow$)
- **Δx vs r_v scatter** (colored by LSI; target-zone overlay)
- **Correlation heatmap** (Δx , r_v , p_r , LSI)
- **Box/violin by profile** (primitive distributions across Figure / MarkMaking / Landscape)
- **Seed-stability view** (variance across seeds for the same prompt)
- **Failure atlas tiles** (top examples per Category-Ledger tag)
- **Systematic Gap table** (prompt $\rightarrow$ acceptance $\rightarrow$ common failure $\rightarrow$ gap note)
- **Color audit rate** (share of rows with an audit per profile)
- **Delta distributions** (`delta_rv`, `delta_dx` hist/violin per profile)
- **Telemetry failure reasons** (bar of `color_status` counts)

Use in studies (how to report, minimal)

- **Profile used** + **gate rule** (≥ 55 and no RED).
- **Acceptance rate**, **band-failure mix**, and **priority-knob distribution**.
- **Seed stability** for repeated prompts.
- **Systematic Gap** notes when language lacks a visual schema (optional but recommended).
- Include a **Color audit** line **only** when telemetry diverges; otherwise omit any mention of color.

Threshold note (footnote-style, don't headline it)

Rule-of-thumb trigger for the audit (engineering guidance):

`|delta_rv|  $\geq 0.15$  or |delta_dx|  $\geq 0.10$` , or gray r_v sits within ± 0.02 of a guard. Keep these thresholds out of scoring; they're advisory.

One-liner (Methods box):

We use LSI-lite, a profiled composition structural exploration that scores off-center gravity (Δx), void ratio (r_v), and rupture energy (p_r) against profile band guards. A weighted geometric mean (ϵ -floored) yields `LSI_lite_100`; an image passes if `LSI_lite_100` ≥ 55 and all bands are OK; edgeless/foregroundless frames are flagged unknown and fail. Profiles: `Figure_Default`, `MarkMaking_Expressive`, `Landscape`; a reflection probe may apply a Δx ROI in `Landscape` only.

Limits

- **Diagnostic, not aesthetic.** “Accepted” $\neq$ “good art.”
- **Profiled, not universal.** Western composition bias; many great images will fail **on purpose**.
- **Scope.** Color, concept, cultural context are out-of-scope; three primitives can't capture all structure.

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Authorship

This framework was architected by Russell Parrish and recursively co-developed inside GPT, Gemini and Claude. Every critique is human-led; every recursion is model-driven. The result: a reasoning layer authored through language, not image manipulation.